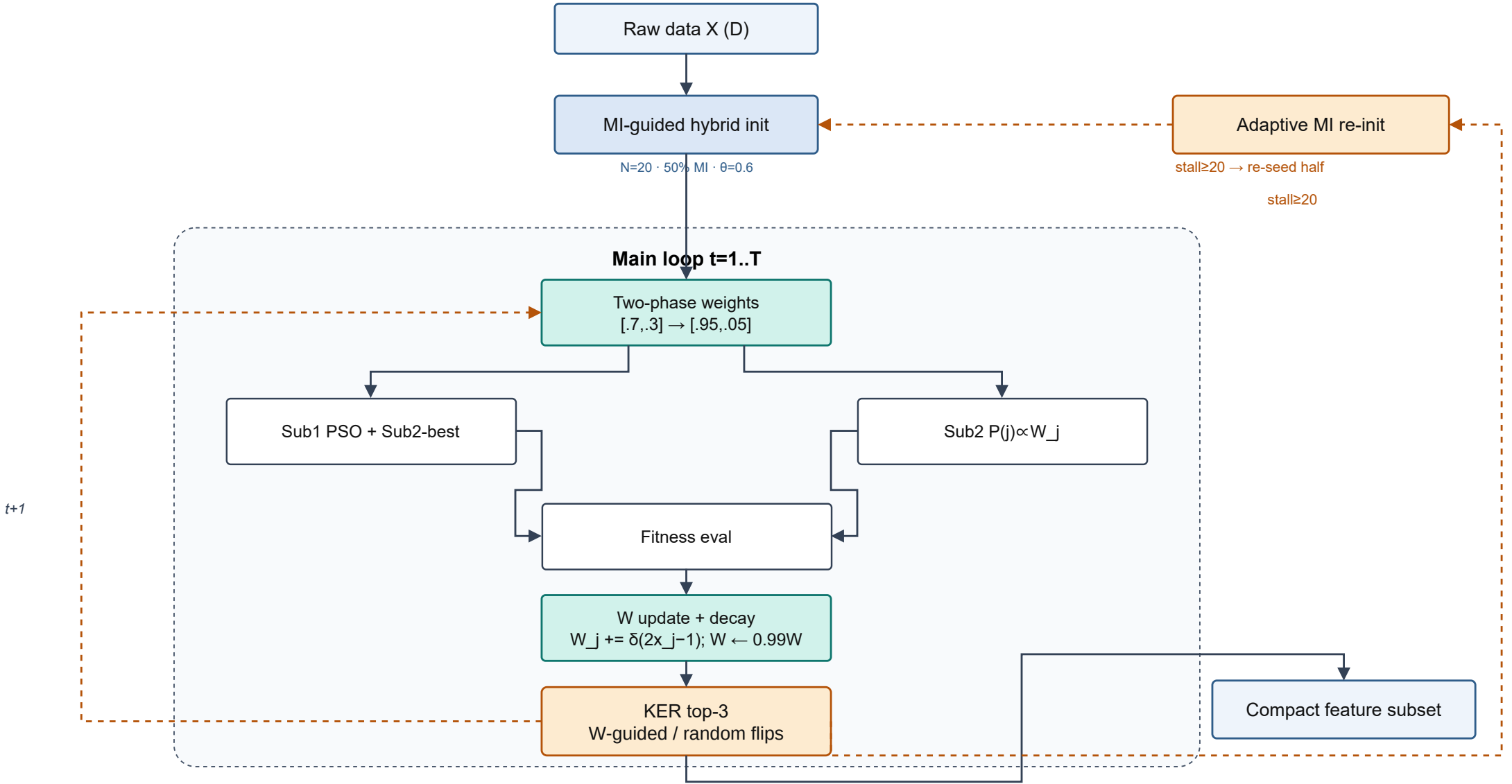

Raw data X (D)
MI-guided hybrid init
N=20 · 50% MI · θ=0.6
Adaptive MI re-init
stall≥20 → re-seed half
stall≥20
Main loop t=1..T
Two-phase weights
[.7,.3] → [.95,.05]
Sub1 PSO + Sub2-best
Sub2 P(j)∝W_j
Fitness eval
W update + decay
W_j += δ(2x_j−1); W ← 0.99W
KER top-3
W-guided / random flips
Compact feature subset
t+1